SUMMER INTERNSHIP REPORT

On

Data Science Master Virtual Internship

(Siemens – AICTE EduSkills Virtual Internship Program, RapidMiner Platform)

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student: 2410030927

Name of Student: DEEPANSHU SHARMA

Section: 3CSE6

Batch: 2026-27

CANDIDATE'S DECLARATION

I, Deepanshu Sharma, Roll No. 2410030927, do hereby declare that the internship report titled “Data Science Master Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature
Student Name: Deepanshu Sharma
Roll No.: 2410030927

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Data Science Master Virtual Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank EduSkills Foundation and Altair Engineering Inc. for designing and conducting this AICTE-approved virtual internship, and Siemens Industry Software for supporting the program. I would also like to thank my faculty coordinator and the Department of Computer Science and Engineering, IILM University, Greater Noida, for their continuous support and encouragement throughout this internship.

I express my gratitude to my parents for their blessings.

Signature
Student Name: Deepanshu Sharma
Roll No.: 2410030927

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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

Data has become one of the most valuable assets for modern organizations, and the ability to extract actionable insight from raw data through data science and machine learning is now a core industry requirement. To help bridge the gap between academic curricula and industry-relevant skills, the All India Council for Technical Education (AICTE), in association with EduSkills Foundation, Altair Engineering Inc., and Siemens Industry Software, offers structured Virtual Internship Programs to engineering students across India.

As part of the 5th semester curriculum requirement of IILM University, Greater Noida, I undertook the 8-week "Data Science Master" Virtual Internship under the AICTE–EduSkills program during April–June 2026. The internship provided structured, week-wise learning on the RapidMiner data science platform, covering the complete data science lifecycle — from business problem identification and data engineering to machine learning, model deployment, and enterprise platform administration.

4.2 Organization Profile

EduSkills Foundation is a Section-8, ISO 9001:2015 and ISO 20000-1:2015 certified, "Great Place To Work" certified not-for-profit organization working under the mission of "Nation Building Through Skills." It partners with AICTE and the National Internship Portal, Ministry of Education, Government of India, to deliver industry-aligned virtual internship programs to engineering students across the country.

The technical content and certification for this internship domain were provided by RapidMiner (an Altair company), a leading enterprise data science and machine learning platform used for data preparation, predictive modelling, and model deployment. The program was supported by Siemens Industry Software (India) Pvt. Ltd., and validated by AICTE, Ministry of Education.

4.3 Problem Statement

Engineering students often graduate with strong theoretical foundations but limited hands-on exposure to industry-grade data science tools and end-to-end machine learning workflows used by real organizations. Most academic exercises stop at model

building in a notebook environment and rarely cover data engineering at scale, enterprise deployment, real-time scoring, or platform administration — skills that recruiters actively look for. The internship was designed to address this gap by training students on a complete, enterprise-grade data science platform (RapidMiner), covering the entire pipeline from raw data to a deployed, monitored model.

4.4 Project Objectives

- To understand how to identify business problems that are suitable for AI and machine learning solutions.
- To learn to access, load, transform, and calculate data using RapidMiner's data engineering tools.
- To build foundational and advanced machine learning models within the RapidMiner Studio environment.
- To master advanced data preparation and automation techniques for large and messy datasets.
- To learn how to deploy and monitor machine learning models in a production-like setting.
- To gain working knowledge of enterprise platform administration, including RapidMiner AI Hub, Radoop, real-time scoring, and the RapidMiner Marketplace.
- To earn industry-recognised RapidMiner certifications validating proficiency at the Professional and Master levels.

4.5 Scope of the Project

The scope of this internship was limited to a structured, curriculum-based virtual learning environment rather than a live client engagement. It covered eight weekly modules progressing from foundational to advanced levels: Applications & Use Cases (Professional and Master), Data Engineering, Machine Learning (Professional and Master), Data Preparation Master, and Platform Administration Master. Each module combined guided learning content with weekly assessments and hands-on exercises inside RapidMiner Studio, culminating in a Final Assessment Test and credential validation. The internship did not involve access to proprietary organizational data; all

exercises were performed on sample/practice datasets provided within the RapidMiner learning environment.

4.6 Technologies and Tools Used

- RapidMiner Studio – visual, code-free workflow designer for data preparation, machine learning, and model validation.
- RapidMiner AI Hub – centralized platform for deploying, scheduling, and managing models and workflows.
- RapidMiner Radoop – extension for running RapidMiner processes on Hadoop/Spark for big-data workloads.
- RapidMiner Real-Time Scoring – module for exposing trained models as real-time scoring web services.
- RapidMiner Marketplace – extension repository for additional operators and connectors.

4.7 System Architecture

The RapidMiner platform follows a modular, pipeline-based architecture that supports the full data science lifecycle. Figure 4.1 summarizes the stages of this pipeline as covered across the internship modules.

Stage	Function
1. Data Access	Connecting to files, databases, and repositories to bring raw data into RapidMiner.
2. Data Engineering	Loading, cleaning, transforming, and calculating data using visual operators.
3. Modelling	Applying machine learning algorithms (classification, regression, clustering) to build predictive models.
4. Validation	Cross-validation and performance evaluation of trained models.
5. Deployment	Publishing models to RapidMiner AI Hub / Real-Time Scoring for consumption by other applications.
6. Administration & Monitoring	Managing users, jobs, and infrastructure via RapidMiner AI Hub and Radoop for enterprise-scale operation.

Figure 4.1: RapidMiner End-to-End Pipeline Architecture

4.8 Methodology

The internship followed a week-wise structured learning methodology as prescribed by the AICTE–EduSkills program, summarised in Table 4.1.

Week	Module	Description

Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Table 4.1: Week-wise Internship Structure

Each module combined self-paced video/text content with a mandatory weekly assessment; select modules additionally required submission of project documentation and assignments to validate hands-on understanding, culminating in a Final Assessment Test in the eighth week.

4.9 Expected Outcomes

- Gained hands-on proficiency in end-to-end data science workflows using the RapidMiner platform.
- Earned two Master-level RapidMiner certifications: "Applications & Use Cases Master" and "Platform Administration Master."
- Successfully completed the 8-week Data Science Master Virtual Internship with an overall Internship Grade of O (Outstanding).
- Developed a practical understanding of enterprise-scale model deployment, monitoring, and platform administration — skills directly applicable to AI/ML and data engineering roles.

4.10 Certificates of Completion and Other Communication Mail Proof

The following pages contain the RapidMiner "Applications & Use Cases Master" certification, the RapidMiner "Platform Administration Master" certification, and the AICTE–EduSkills Internship Offer Letter, as proof of completion and communication for this internship.

RapidMiner – Applications & Use Cases Master Certification



RapidMiner – Platform Administration Master Certification



AICTE–EduSkills Internship Offer Letter

EduSkills
Nation Building Through Skills

Ref No: C16Y2026M061412778

Internship Offer Letter

Student Name	: Deepanshu Sharma
Stream	: B.Tech
Branch	: B.Tech(Computer Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Siemens Data Science Master
Internship Duration	: April 2026 - June 2026
AICTE Student ID	: STU699effc8d7f5a1772027848

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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IP: 13.204.68.82 | Device: Desktop

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The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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CHAPTER 5: BIBLIOGRAPHY / REFERENCES

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